ChatGPT

Golden Spirals in Geometry and on Curved Surfaces

Historical Geometry of the “Golden” Spiral

Ancient Foundations – Golden Ratio and Early Spirals: The concept of the golden ratio (approximately 1.618, often denoted φ) was well known in classical Greek geometry. Euclid’s *Elements* defines dividing a line in “extreme and mean ratio,” which essentially describes the golden ratio [1]. Pythagorean scholars also studied this ratio, observing it in the pentagon and pentagram; evidence suggests the Pythagoreans understood that the diagonals of a regular pentagon intersect in golden ratio proportions [2]. However, while Greek geometers like Euclid and Pythagoras revered geometrical ratios, there’s no record that they explicitly described a **spiral** based on the golden ratio. Early Greek mathematics did explore other spirals – most notably, Archimedes in the 3rd century BCE studied the **Archimedean spiral**, a curve with constant separation between its turns [3]. In his treatise *On Spirals*, Archimedes defined this spiral (rising from the pole at uniform speed while the radius rotates uniformly) and even used it to square the circle [4] [3]. This Archimedean spiral is a purely geometric curve (polar equation $r = a \cdot \theta$) and was one of the first rigorous treatments of a spiral in Western mathematics. It does not involve the golden ratio, but its early study set the stage for later spiral discoveries. Other classical examples include the Spiral of Theodorus (composed of sequential right triangles), though again these lacked any golden-ratio connection. In summary, antiquity provided a foundation in both the golden ratio and in spiral curves separately – the **fusion** of the two concepts into a “golden spiral” had to await much later developments.

Logarithmic (Equiangular) Spiral Emergence: In the 17th century, the study of spirals took a major leap with the discovery of the **logarithmic spiral**, a curve that does embody self-similar growth – of which the golden spiral is a special case. René Descartes is credited with first analyzing the equiangular spiral in 1638 [5]. This spiral has the defining property that it cuts all radial lines at a constant angle (hence “equiangular”), and can be described in polar form by *r = a·e^{bθ}*. Descartes’ work was soon extended by others; Evangelista Torricelli, for example, found the length of a logarithmic spiral curve analytically [6]. The logarithmic spiral captivated many mathematicians because of its marvellous self-similarity. In 1692, Jakob Bernoulli studied it extensively and dubbed it *spira mirabilis*, the “miracle spiral,” due to its property of reproducing its shape under scaling [5]. Bernoulli was so enchanted that he requested a logarithmic spiral be engraved on his tomb, calling it “the marvelous spiral” that symbolically resurges upon itself [7]. (Indeed, the equiangular spiral curve was carved on his tombstone in Basel [7].) This logarithmic spiral is the general form that includes the **golden spiral** as one instance. A **golden spiral** is simply a logarithmic spiral whose growth factor per quarter-turn is φ, the golden ratio [8]. In other words, for every 90° of rotation, the radius of a golden spiral increases by a factor of 1.618, yielding a distinctive tightness of coil. Notably, while the golden spiral was not singled out by the early geometers, it can be constructed geometrically as an approximation: for example, by inscribing quarter-circle arcs in an ever-expanding sequence of golden rectangles (whose side lengths follow Fibonacci numbers). Johannes Kepler in 1608 had already observed the link between the Fibonacci sequence and φ (each term’s ratio to the previous approaches 1.618) [9], and later mathematicians realized these Fibonacci-based quarter-circle spirals approximate the true golden spiral. The figure below illustrates a classic construction of a golden spiral via contiguous golden rectangles:

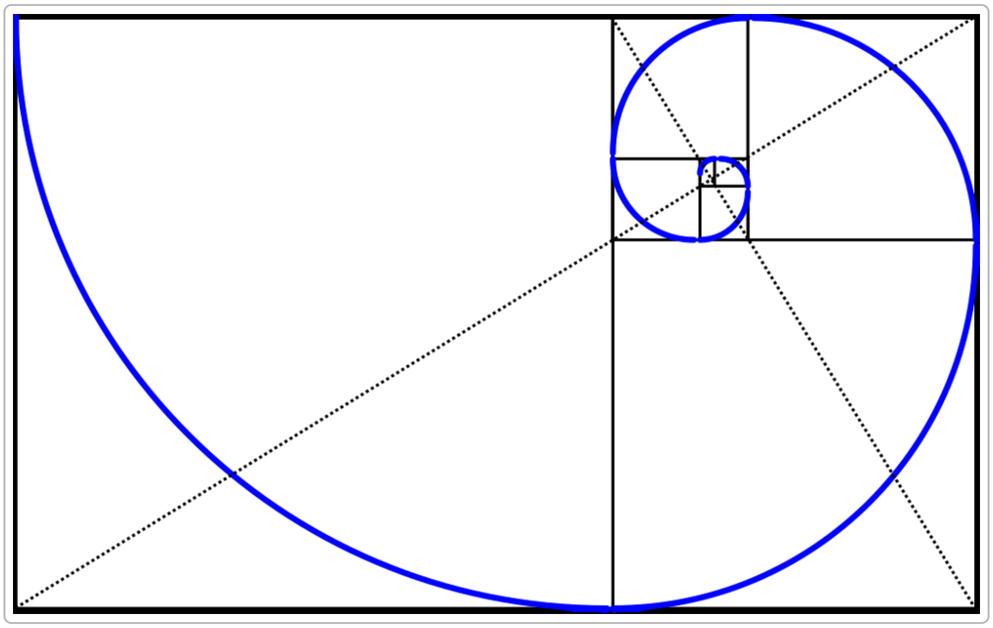

Figure: A classical golden spiral (blue curve) constructed by drawing circular arcs through opposite corners of subdivided golden rectangles. At each quarter-turn, the spiral's radius grows by the factor φ, producing a self-similar curve. [8]

With the rise of analytic methods in the Enlightenment, the properties of logarithmic spirals (including the golden spiral) became well understood. One remarkable property is self-similarity: any portion of a logarithmic spiral, when magnified, is congruent to the whole – a fact that Bernoulli treasured. Mathematically, Jakob Bernoulli and his brother Johann showed that the logarithmic spiral is invariant under certain transformations; for instance, the evolute (curve of curvature centers) and involute of an equiangular spiral are identical to the original spiral [10]. This deep property made the curve a natural object of study in the new calculus of the 18th century. **Leonhard Euler** and others embraced the analytic description: Euler wrote the polar equation of the logarithmic spiral as $r = Ce^{k\theta}$ and could derive quantities like curvature from it. In fact, the term "logarithmic spiral" arises because taking the logarithm of r yields a linear function of θ. While Euler did not specifically single out the golden-growth case in his writings, his work provided the general formulas that include it. The golden spiral, being one specific equiangular spiral, inherits all these nice properties (constant angle, self-similarity, segments along any ray forming a geometric progression [11]). It's worth noting that the *aesthetic* appeal of φ was also hitting a peak during this period – for example, 18th-century scholars like Martin Ohm later coined the term "golden section," and art theorists praised the golden ratio – but here we focus strictly on geometry. In geometry, the golden spiral stood out as a beautiful curve exemplifying the unity of φ and the logarithmic form.

Spirals on Curved Surfaces – Spherical Geometry in the Enlightenment: By the 18th and early 19th centuries, the scope of geometry expanded beyond the flat plane. Mathematicians like Euler and Gauss began examining curves on curved surfaces, laying groundwork for non-Euclidean geometry. A prominent spiral-like curve in spherical geometry is the **loxodrome** (or rhumb line) – the path on a sphere that crosses all meridians at the same constant angle. This is essentially the spherical analogue of an equiangular spiral:

on a globe, a loxodrome spirals infinitely toward a pole while maintaining a fixed bearing (constant angle β to the north-south meridians). The concept had arisen from navigation: Portuguese cosmographer Pedro Nunes identified the rhumb line in 1537 as distinct from a great circle route [12], and Gerardus Mercator's 1569 world map famously maps loxodromes to straight lines [13]. Enlightenment mathematicians brought rigor to these ideas. Euler in the 18th century studied problems of navigation and cartography; although Mercator's projection was known, Euler explored the calculus of variations on spherical surfaces, which includes deriving the geodesics (shortest paths) and could address curves like rhumb lines. (Euler's work on great-circle sailing and related integrals helped establish spherical trigonometry formulas, though the full parametric equation of a loxodrome on a sphere was formally written a bit later [14].) Notably, the loxodrome on a sphere *is* a form of logarithmic spiral when projected appropriately: in 1905, mathematician H. Nobel showed that stereographic projection of a spherical loxodrome yields a logarithmic spiral in the plane [15]. This means the same "miraculous spiral" Bernoulli loved appears when one views a constant-bearing spherical path on a flat map (except Mercator's special map, where it becomes a straight line). Thus, even though Euler and Gauss did not speak of a "golden spiral" on a sphere, they were aware of the general phenomenon of spiral paths on curved surfaces. **Carl Friedrich Gauss**, in his monumental 1827 work on differential geometry (*Disquisitiones Generales circa Superficies Curvas*), developed tools for understanding curves on surfaces of any curvature. Gauss studied Earth's shape (an oblate spheroid) and the behavior of lines on it as part of geodesy. His theorem *egregium* and the Gaussian curvature concept allowed mathematicians to analyze how a curve like a loxodrome or geodesic behaves on an ellipsoid versus on a sphere. For instance, on an ellipsoid (Earth), a constant compass bearing path is no longer a simple analytic spiral, but Gauss's methods (and Clairaut's earlier formula for geodesics) gave insight into such routes. In summary, by the end of the Enlightenment, Western mathematicians had a firm grasp on spirals both in the plane (e.g. logarithmic and golden spirals) and on curved surfaces (e.g. spherical spirals like rhumb lines). The stage was set to appreciate these curves not just as navigational oddities but as mathematical objects connecting geometry, analysis, and even natural phenomena.

Recreational and Aesthetic Mathematics – Spirals on Curved Surfaces

Mathematicians and math enthusiasts have long been fascinated by spiral patterns, and this extends to **spirals drawn on curved surfaces** like spheres and ellipsoids. Below we compile a variety of visually compelling and intellectually stimulating explorations – from art and nature-inspired constructions to puzzles and creative challenges – all involving spirals (including golden or logarithmic spirals) on curved surfaces.

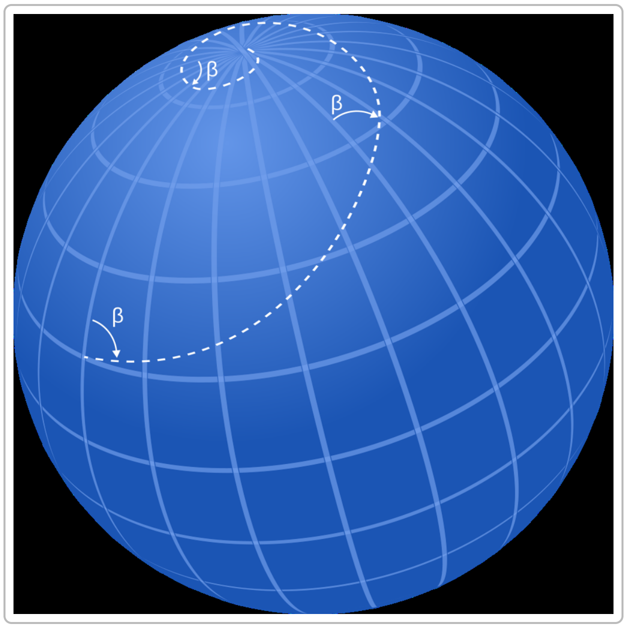

Figure: A loxodromic spiral (constant-bearing rhumb line) on a sphere. The path (dashed) makes a fixed angle β with all meridians and winds infinitely toward the pole. In navigation this is the route of constant compass heading. Such spherical spirals are equiangular spirals mapped onto a sphere – their projection from the pole yields a logarithmic spiral in the plane [15] *.*

- **Escher's Spherical Spirals – Art Meets Mathematics:** One of the most famous aesthetic explorations of spherical spirals comes from the artist M.C. Escher. In his 1958 woodcut *Sphere Surface with Fish*, Escher arranged interlocking fish along eight spiral paths that wrap around a sphere from pole to pole [16] [17] . Remarkably, these paths are *loxodromes* – each fish follows a constant-angle spiral towards the poles, a tribute to the mathematical idea of a spherical logarithmic spiral. Escher was deliberate in using such curves; another piece, *Sphere Spirals*, also employs symmetric loxodromic curves on a sphere's surface [18] . These artworks are visually striking: the spirals ensure that the pattern of fish (or other motifs) repeat self-similarly as they shrink toward the poles, much like a plane logarithmic spiral would. Math enthusiasts enjoy Escher's work not only for its beauty but because it vividly demonstrates a complex geometrical concept in an intuitive way. The fact that the fish never meet at the poles (they get arbitrarily small, looping infinitely many times) illustrates the infinite nature of the spherical spiral [19] [18] . Escher's sphere art has inspired many – for example, mathematicians have analyzed how accurately his spirals match true loxodromes and even recreated his patterns with different figures. This fusion of art and math is a **classic** example of recreational mathematics: it invites viewers to appreciate geometry on a curved canvas.

- **Phyllotaxis on a Sphere – Fibonacci Spirals Wrap the Globe:** Spirals are abundant in nature, often related to the golden ratio – sunflower seed heads and pinecones exhibit opposing families of Fibonacci-number spirals. A natural question is: can we extend this idea to a spherical surface? It turns out there is a well-loved construction known as the **Fibonacci sphere** (or "golden spiral sphere") used to distribute points nearly uniformly on a sphere. The idea is to place points on the sphere at increments of the golden angle (≈137.5°) in longitude, while spacing them evenly in latitude. The result is that points fall along a spiral path that winds from the south pole to north pole. This algorithm, inspired by plant phyllotaxis, produces an extremely uniform and aesthetically pleasing arrangement of points [20]. In effect, one is mapping a flat Fibonacci lattice onto the sphere's surface via a spiral-like sequence [21]. Math and computer graphics enthusiasts use this for tasks like picking well-spaced sample directions. The construction is elegant: as one moves up the sphere, each new point is rotated by the golden ratio around the sphere from the previous, so no longitudinal alignment ever repeats [20]. The end result is a beautiful double spiral grid on the sphere, analogous to the criss-crossing spiral patterns on a sunflower, but now wrapping a full globe. This **visualization** appeals to the aesthetic sense (the points form spiraling "stripes" if connected) and also highlights the ubiquity of φ in optimal packing problems. It's an example of a simple, fun exercise with deep connections – students can try coding the Fibonacci sphere and watch the spirals emerge, bridging coding, math, and art.

- **Puzzles and Challenges with Spherical Spirals:** Spirals on spheres also lead to intriguing mathematical puzzles. For instance, one classic challenge asks: *Can two different loxodromic spirals on a sphere never intersect?* Surprisingly, the answer is yes – if you choose the same constant angle and start them at carefully chosen offset longitudes, two such spirals can wind from pole to pole without ever crossing each other [22]. Math enthusiasts have explored this problem of "non-intersecting loxodromes," finding, for example, that at a 60° constant bearing you can fit exactly two spirals that never meet [22]. This is a delightfully counterintuitive result since one might expect multiple spirals on a sphere to eventually intersect. Another recreational problem involves measuring spiral paths: *If a ship follows a constant compass bearing, how many loops around the world will it make by the time it reaches the pole?* The answer is that it will circle infinitely many times as it asymptotically approaches the pole, but one can compute how the spacing of each loop decreases geometrically. For a given angle β, the "loop spacing" in latitude is constant, leading to an infinite series. This ties into another fun fact: the total length of a loxodrome from equator to pole is finite despite circling infinitely often (a spherical analog of the paradox of Achilles and the tortoise) [5] [23]. Such problems are enjoyable exercises in spherical trigonometry and analysis, accessible to an enthusiastic student with a bit of guidance. They capture the imagination by showing how a simple rule (constant angle) yields a path with endless twists. Additionally, one can turn the tables and create a **treasure-hunt puzzle**: "Starting at point X on Earth, walk always heading northwest (at say 45° to the meridians). After traveling a very long distance, you arrive at point Y. Where is Y relative to the North Pole, and how far have you walked?" – The solver would discover Y is arbitrarily close to the pole and that the path length can be calculated by an integral formula. Such puzzles encourage understanding the geometry of the sphere in a concrete way.

- **Creative Constructions on Ellipsoids:** Fewer classic problems exist for ellipsoids (spheroids) specifically, but we can invent analogues. An ellipsoid (like a flattened sphere) adds complexity – the curvature isn't uniform. A compelling open-ended challenge for enthusiasts is to **design a spiral on an ellipsoid**. For example: *find a path on an ellipsoidal globe that maintains a constant angle with the lines of longitude.* On a perfect sphere this is the loxodrome; on an ellipsoid, it will not be a simple logarithmic spiral, but one could attempt numerical plotting. This becomes a mix of differential geometry and computation – a "DIY research" activity. Another idea is inspired by art: imagine wrapping a golden spiral around a rugby-ball-shaped ellipsoid such that it is perfectly symmetric end-to-end. What would its equation look like? Could one 3D-print an ellipsoid with a golden spiral groove carved in it? Challenges like these have a recreational flavor despite being unexplored: they invite creative thinking and can lead to visual **experiments**. Even without formulas, a hobbyist might take a prolate spheroid and mark off equal angles along meridians, connecting them in a smooth curve to approximate a spiral. The result could be a unique decorative object (an "ellipsoidal

Fibonacci pumpkin" perhaps!). Such creative explorations carry the spirit of recreational mathematics – they are *just for fun*, yet they deepen one's intuition about geometry on curved surfaces.

In conclusion, golden spirals and their kin not only hold an honored place in the history of geometry but also continue to spark joy through creative play. From the strictly geometric treatments by Euclid, Archimedes, and Bernoulli to the playful visualizations of Escher and the elegant spirals in nature, these curves bridge the gap between serious mathematics and recreational exploration. Whether drawn with straightedge and compass on a flat page or traced along the curve of a globe, spiral patterns provide endless fascination – a meeting point of beauty, puzzle, and mathematical truth.

Sources: Euclid's definition of the golden ratio [1]; Pythagorean study of the pentagon [2]; Archimedes' **On Spirals** treatise [3]; Descartes' and Bernoulli's work on equiangular spirals [5]; the golden spiral's growth factor definition [8]; spherical spiral (loxodrome) characteristics [19] [15]; Escher's use of loxodromic spirals in art [18] [16]; Fibonacci spiral distribution on spheres [21] [20]; and classic loxodrome puzzles discussed by mathematicians [22].

---

[1] [2] Golden ratio - MacTutor History of Mathematics
https://mathshistory.st-andrews.ac.uk/HistTopics/Golden_ratio/

[3] Archimedean spiral - Wikipedia
https://en.wikipedia.org/wiki/Archimedean_spiral

[4] On Spirals - Wikipedia
https://en.wikipedia.org/wiki/On_Spirals

[5] [6] [7] [10] [11] [23] Equiangular Spiral - MacTutor History of Mathematics
https://mathshistory.st-andrews.ac.uk/Curves/Equiangular/

[8] Golden spiral - Wikipedia
https://en.wikipedia.org/wiki/Golden_spiral

[9] StoryAlity #52 – The Golden Ratio / The Golden Spiral / The Fibonacci Sequence | StoryAlity
https://storyality.wordpress.com/2012/12/23/storyality-50-3-the-golden-ratio-the-golden-spiral-the-fibonacci-sequence/

[12] [13] [14] [15] Title
https://dergipark.org.tr/tr/download/article-file/2595623

[16] Loxodromic Spirals in M. C. Escher's Sphere Surface
https://scholarship.claremont.edu/jhm/vol4/iss2/4/

[17] Photos - Home page Tis Veugen
http://tistis.nl/photos3.html

[18] [19] Spherical Spiral -- from Wolfram MathWorld
https://mathworld.wolfram.com/SphericalSpiral.html

[20] [21] How to evenly distribute points on a sphere more effectively than the canonical Fibonacci Lattice - Extreme Learning
https://extremelearning.com.au/how-to-evenly-distribute-points-on-a-sphere-more-effectively-than-the-canonical-fibonacci-lattice/

[22] geometry - On nonintersecting loxodromes - Mathematics Stack Exchange
https://math.stackexchange.com/questions/15801/on-nonintersecting-loxodromes